

Determinants of Gold Price Volatility in Response to Non-Farm Payrolls Releases: An Econometric Analysis

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Abstract

This study analyzes the determinants of gold price volatility in the five minutes before and after the release of Non-Farm Payrolls (NFP), using an econometric approach. NFP represents a key indicator of the U.S. labor market, released monthly by the Bureau of Labor Statistics, and is often considered a catalyst for movements in financial markets. Volatility is modeled as a function of three main variables: the weekly average volatility, the deviation between forecasted and actual NFP values, and the presence of significant geopolitical events, identified through a dummy variable based on interquantile analysis. The initial model highlighted the presence of autocorrelation in the residuals, which was tackled through a model transformation. The results reveal some interesting dynamics in gold price movements in response to NFP releases, suggesting useful insights for further investigation. This study represents a first step toward a more detailed understanding of the interactions between financial markets and macroeconomic events of global significance.

1 Introduction

The gold market has always been considered a safe-haven asset, attractive in times of economic and geopolitical uncertainty. This study focuses on the analysis of the financial instrument XAUUSD, a spot contract that replicates the price of gold in US dollars. XAUUSD stands out for its high liquidity and significant daily movements, making it particularly suitable not only for intraday trading but also for the implementation of quantitative strategies. Unlike gold futures, which have specific expiration dates and rollover costs, XAUUSD is a flexible instrument that allows traders to trade gold in real-time without expiration constraints, accurately replicating the value of physical gold. These characteristics make it one of the most popular instruments for trading in global markets. A key moment for the gold market, and consequently for XAUUSD, is the release of Non-Farm Payrolls (NFP), an

economic report published on the first Friday of each month by the U.S. Bureau of Labor Statistics. This indicator measures the change in non-farm employment compared to the previous month and is one of the most closely watched metrics for assessing the health of the U.S. economy. NFP influences market expectations regarding monetary policy, as a significant deviation from forecasts could prompt the Federal Reserve to adjust interest rates. For this reason, the NFP release day is highly anticipated by traders, as it can trigger substantial volatility in financial markets, including gold. Although volatility is an intrinsic characteristic of the gold market, the price behavior of XAUUSD during NFP releases is not uniform. In some cases, significant price movements occur in the minutes following the release, whereas in others, the market reaction is more subdued. This study aims to analyze the dynamics that influence XAUUSD volatility in the five minutes before and after the NFP release, seeking to understand the conditions that lead to a significant increase in volatility or, conversely, to a limited market response. The econometric regression used in this study considers the price range of XAUUSD during this time window as the dependent variable. The independent variables include:

- **Weekly average volatility:** Represents the overall level of market activity in the week preceding the release.
- **Deviation between forecasted and actual NFP values:** Captures the magnitude of the information shock to the market, based on the comparison between analysts' forecasts collected from platforms such as Investing.com and the actual value released by the U.S. Bureau of Labor Statistics.
- **Presence of significant geopolitical events:** Modeled through a dummy variable with "Yes" or "No" attributes, indicating the presence or absence of a relevant geopolitical event on the day of the NFP release. This variable was constructed based on global media resonance.

During the analysis, autocorrelation was detected in the model's residuals (Durbin-Watson test), which was addressed through a linear transformation of the model to ensure residual independence and estimate efficiency. The study results indicate that while weekly average volatility and the presence of significant geopolitical events appear to have a potential impact on price volatility, none of these variables exhibited a statistically significant effect. Additionally, the deviation between forecasted and actual NFP values does not seem to play a crucial role. These findings provide useful insights for developing more comprehensive quantitative models.

2 Literature Review

Gold is one of the most studied assets in financial markets, both for its historical role as a safe-haven asset and for its relevance as a speculative instrument. The volatility of gold prices has been analyzed in various contexts, with a particular focus on the effects of macroeconomic and geopolitical events, as well as market dynamics that emerge during periods of high uncertainty. From a practical perspective, gold also serves as a key instrument for developing quantitative strategies, thanks to its liquidity and frequent price movements, making it particularly appealing for those seeking to build portfolios based on automated models.

Among the most significant macroeconomic events for financial markets, Non-Farm Payrolls (NFP) are among the most closely watched economic releases due to their influence on monetary policy expectations and global market behavior. NFP provides a monthly measure of non-agricultural employment in the United States, published by the Bureau of Labor Statistics. Their ability to generate immediate volatility in financial markets makes them an ideal context for analyzing short-term reactions, such as those in the gold market.

This study originates from a finance student's interest in intraday market dynamics and the desire to apply academic knowledge such as econometrics and coding to practical problems. The goal is to examine price fluctuations in the financial instrument XAUUSD, a spot replica of gold prices, within the five minutes before and after the release of NFP. While it does not aim to propose significant innovations, this work seeks to develop a model that can be used to explore key factors influencing intraday gold price movements, laying the groundwork for future strategies.

The literature provides valuable insights into the effects of macroeconomic news on markets, but the specific analysis of gold around NFP releases, with a focus on short time horizons, remains relatively underexplored. This study integrates variables such as the weekly average volatility, the deviation between forecasted and actual NFP values, and the presence of relevant geopolitical events to better understand these dynamics. This work serves as a practical exercise in integrating theoretical and quantitative tools into the construction of econometric models, with potential applications for automated trading strategies in future professional contexts.

3 Methodology

3.1 Dataset Construction

The construction of the final dataset, used for the econometric analysis, was a central phase of this work. It was necessary to integrate data from various sources, transform

them, and combine them to obtain a coherent and complete dataset suitable for performing regression analysis. The key steps are described below.

3.1.1 Data Sources

1. XAUUSD Prices:

- **Source:** TickStory (Dukascopy database).
- **Collected Data:**
 - Minute candles: from October 1, 2020, to November 2, 2024.
 - Daily candles: same time range.
 - **Time format:** UTC.

2. Non-Farm Payrolls (NFP) Data:

- **Source:** Investing.com (Bureau of Labor Statistics).
- **Available period:** from 2 October 2020 to 4 October 2024.
- **Collected Data:** Actual values, forecasted values, deviations.

3. Geopolitical Events:

- **Source:** Media Cloud.
- **Available period:** from 3 January 2021 to 4 October 2024.
- **Usage:** construction of a dummy indicating the presence of relevant geopolitical events, based on criteria described in detail in the dedicated section.

3.2 Dataset construction phases:

1. Preparation of Daily Data:

(a) Creation of the file `data_Daily` starting from three main sources:

i. XAUUSD Daily Prices:

- Calculation of the **daily** range: (High - Low) for each day.
- Addition of the column **dayofweek** to identify the day of the week (with a focus on Fridays, the NFP release day).

ii. Geopolitical Events:

- Transformation into a file with three columns: date, media resonance percentage, and dummy variable ("Yes"/"No").

iii. NFP Data:

- Importation of actual, forecasted, and deviation values for each NFP release, associated with release Fridays.

(b) **Data Merging and Manipulation in Excel:**

- The three sources were merged to create the initial dataset, later imported into R as **data_Daily**.

2. **Calculation of weekly average volatility:**

- The column **volatility_mean_week** was added, computed as the cumulative average of the daily range from the start of the week.

3. **Time zone adjustments:**

- NFP data releases always occur at 8:30 am Eastern Time (ET). A column was added to represent this time, later converted to UTC to align with XAUUSD intraday data.

4. **Creation of the "filtered_data" subset:**

- Rows with missing values were removed.
- The dataset was reduced to 44 observations (Fridays with NFP releases).

5. **Importation and integration of intraday data:**

- XAUUSD intraday data (1,446,574 observations) were filtered to match the relevant dates.
- The result, **merged_data**, contains all intraday data for NFP Fridays.

6. **Isolation of relevant time window:**

- Intraday data were filtered to keep only the 5 minutes before and after the NFP release.

7. **Final dataset creation:**

- The dependent variable **price_range** was computed as the difference between max and min prices in the 10-minute window.
- Inclusion of independent variables: **volatility_mean_week**, geopolitical dummy, and expected-actual variance
- The final dataset **volatility_news** contains 44 observations ready for regression.

This process, although complex, was essential to guarantee the quality and consistency of the dataset, laying the foundations for the econometric analysis described in the following sections.

3.3 Creation of the Dummy Variable for Geopolitical Events

3.3.1 Data Source

The data used to identify the presence of significant geopolitical events were extracted from Media Cloud, a platform that analyzes media resonance in journalistic sources. The reference period spans from January 1, 2020, to October 29, 2024. The CSV file obtained contains two main columns:

- **Date:** The date of the event.
- **News Resonance (%):** The percentage of articles published in major U.S. media corresponding to specific keywords associated with significant geopolitical events.

To identify significant events capable of influencing gold volatility, a list of keywords was developed across macro-categories, each including terms selected for their relevance in the econometric analysis.

3.3.2 Keyword Categories

1. Geopolitical Events

- *"Geopolitical conflict"*: Identifies international conflicts raising global uncertainty and reinforcing demand for safe-haven assets.
- *"Economic sanctions"*: Captures the impact of economic sanctions that destabilize economies and markets.

2. Economic and Financial Events

- *"Central bank rate decision"*: Monitors unexpected interest rate decisions affecting the opportunity cost of holding gold.
- *"Equity market crash"*: Indicates periods of crisis in stock markets, increasing gold's attractiveness.

3. Global Events

- *"Pandemic outbreak"*: Detects major health events, such as COVID-19, impacting financial markets.
- *"Natural disaster"*: Captures the economic impact of extreme natural events, destabilizing global supply chains.

4. Unexpected Policy Changes

- *"Monetary stimulus"*: Monitors sudden central bank interventions affecting currency and gold values.
- *"Tariff hike"*: Identifies unexpected increases in trade tariffs, with potential global market repercussions.

5. Market and Trade Events

- *"Gold demand spike"*: Signals sudden increases in gold demand, often linked to crises or uncertainty.
- *"Commodity price shock"*: Detects abrupt fluctuations in the commodity market, influencing gold prices.

3.3.3 Objective

The goal was to create a dummy variable (**anomaly**) indicating whether, on a given day, a significant geopolitical event was present (**Yes**) or not (**No**).

3.3.4 Methodology

1. Distribution Analysis:

- To determine a significant resonance threshold, a density estimation of the variable **news_resonance** was performed.
- The density function showed a ****right-skewed distribution****, with a long right tail indicating days with exceptionally high media resonance.

2. Statistical Approach: Interquartile Range (IQR):

- Due to the skewed distribution, a standard normal-based approach was deemed inappropriate.
- Instead, the ****Interquartile Range (IQR)**** method was applied:

$$IQR = Q_3 - Q_1 \tag{1}$$

where:

- Q_1 = first quartile (25% of the data)
- Q_3 = third quartile (75% of the data)

3. Natural Identification of Anomalous Thresholds:

- Using the IQR method, anomaly thresholds were defined based on Q_3 and optionally $Q_3 + 1.5 \times IQR$:

- Q_3 : Values in the 75th percentile are already considered ****unusual events****, as only 25% of the data exceed this threshold.
- $Q_3 + 1.5 \times IQR$: Frequently used for detecting ****extreme outliers****, useful for focusing on the most exceptional events.

4. Relevance Threshold:

- After exploratory analysis, a simple and direct threshold was chosen for identifying relevant events.
- All values above Q_3 were classified as ****Yes****, as they represent days with media resonance above the 75th percentile, indicative of significant events.

3.3.5 Data visualization for Dummy

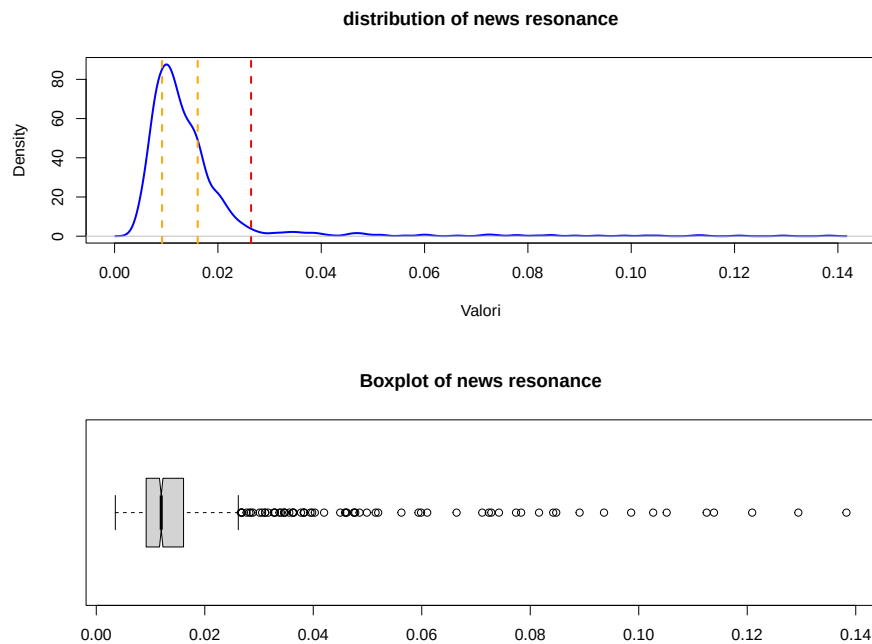


Figure 1: Distribution and Boxplot of News Resonance

3.3.6 Implementation in R

The following code shows the process of calculation and categorization:

```

1 # Loading the data
2 data <- read.csv('distribution_world_Event.csv', header = TRUE)
3
4 # Calculation of quartiles
5 Q1 <- quantile(data$news_resonance, 0.25)

```



```

6 Q3 <- quantile(data$news_resonance, 0.75)
7
8 # Creation of the dummy variable
9 data$anomaly <- ifelse(data$news_resonance > Q3, "Yes", "No")
10
11 # Exporting the updated file
12 write.csv(data, 'my_dataframe.csv', row.names = FALSE)

```

3.4 Econometric Model Description

The objective of this study is to analyze the factors determining intraday gold volatility during the release of Non-Farm Payrolls (NFP), using an econometric approach. The chosen dependent variable is `price_range`, calculated as the difference between the highest and lowest price of XAUUSD in the five minutes before and after the NFP release.

The econometric model considers three main explanatory variables:

1. **Weekly Average Volatility (`volatility_mean_week`):** This variable represents the average volatility of the current week, calculated as the cumulative mean of the daily ranges up to the release day. The objective is to verify whether a week characterized by high volatility affects the market's reaction to NFP. In particular, this study investigates whether the market, after a week of elevated volatility, maintains strong reactivity or instead "ignores" the NFP release due to an attention absorption phenomenon.
2. **Dummy Variable for Geopolitical Events (`dummy_geopolitics`):** This binary variable indicates the presence or absence of significant geopolitical events on the NFP release day. It was constructed using media resonance data provided by Media Cloud, based on targeted keywords to identify high-impact events.
 - The keywords were selected based on their relevance to gold markets and are categorized into five main groups:
 - **Geopolitical Events:** e.g., *"Geopolitical tension," "Military conflict," "Sanctions."*
 - **Economic and Financial Events:** e.g., *"Interest rate hike," "Stock market crash," "Currency crisis."*
 - **Global Events:** e.g., *"Pandemic outbreak," "Natural disaster," "Cyberattack on financial institutions."*
 - **Unexpected Policy Changes:** e.g., *"Government stimulus," "Trade war escalation."*

– **Market Events:** e.g., "*Gold buying surge*," "*Safe-haven asset spike*," "*Commodities rally*."

- Media Cloud provided data through the *Attention Over Time* metric, which measures the number of journalistic articles corresponding to selected queries. The dummy variable was then constructed by classifying days with media resonance above the 75th percentile as significant event days ("Yes"), while all others were classified as "No."

3. **Forecast Error (`forecast_error`):** This variable measures the difference between the actual NFP value and the market consensus obtained from Investing.com. The goal is to assess whether the "informational shock" derived from a significant discrepancy between the forecasted and actual value influences intraday gold volatility. A high differential might suggest that market participants did not fully anticipate the impact of the real data.

The econometric model is represented by the following equation:

$$\text{price_range}_t = \beta_0 + \beta_1 \cdot \text{volatility_mean_week}_t + \beta_2 \cdot \text{dummy_geopolitics}_t + \beta_3 \cdot \text{forecast_error}_t + \varepsilon_t \quad (2)$$

where:

- price_range_t : XAUUSD price range in the 5 minutes surrounding the NFP release (dependent variable).
- $\text{volatility_mean_week}_t$: Weekly average volatility.
- $\text{dummy_geopolitics}_t$: Binary variable indicating the presence of significant geopolitical events.
- forecast_error_t : Difference between forecasted and actual NFP values.
- ε_t : Error term.

This model was chosen to explore the market reaction dynamics of gold to NFP releases, considering both pre-existing volatility factors and external variables (geopolitical events and forecast deviations) that may influence market response.

4 Data Analysis

The initial model estimation was performed using **Ordinary Least Squares (OLS)** through the `lm()` function in R. This estimation assumes that the residuals (ε) satisfy all fundamental OLS assumptions, including the independence of errors.

To verify this crucial assumption, the **Durbin-Watson test** was applied, which serves to diagnose the presence of autocorrelation in residuals. The test statistic is computed as follows:

$$d = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} \quad (3)$$

where e_t represents the estimated residuals. The test statistic d ranges between 0 and 4:

- $d \approx 2$: absence of autocorrelation.
- $d < 2$: positive autocorrelation.
- $d > 2$: negative autocorrelation.

4.1 Durbin-Watson Test Results

In this case study, the computed value of the statistic was $d = 1.491955$ with a *p-value* of 0.06.

These results are interpreted as follows:

4.1.1 Comparison with Critical Thresholds

Based on the number of observations ($n = 44$) and independent variables ($k = 3$), the estimated critical thresholds are:

- $d_L = 1.39$ (lower bound).
- $d_U = 1.65$ (upper bound).

Since $d_L < d < d_U$, the result falls within the **indeterminate zone**, where it is not possible to draw definitive conclusions about the presence or absence of autocorrelation.

4.1.2 Analysis of the *p-value*

- The *p-value* of 0.06 is slightly above the significance level $\alpha = 0.05$, indicating that there is no statistically sufficient evidence to reject the null hypothesis of no autocorrelation.

These results suggest that, although strong evidence of autocorrelation is lacking, the presence of **mild positive autocorrelation** cannot be ruled out.

4.2 Implications of Residual Autocorrelation

The presence of autocorrelation in residuals can have significant consequences for the validity of estimations:

1. **Coefficient estimates remain unbiased:**

- Even in the presence of autocorrelation, OLS produces estimates that are correct on average ($E[\hat{\beta}] = \beta$).

2. **Estimates are not efficient:**

- The estimators do not have the minimum possible variance, violating the Gauss-Markov Theorem.

3. **Statistical inference is unreliable:**

- The variances of estimators may be underestimated, leading to errors in significance tests and confidence intervals.

4.3 Conclusion

Although the test did not provide sufficient statistical evidence to reject the null hypothesis of no autocorrelation, the value of d close to the critical zone and the p -value near 0.05 suggest the possibility of mild positive autocorrelation. To ensure model robustness, a transformation was applied to correct for autocorrelation, as described in the next section.

5 Transformed Model

5.1 Estimation of the Autocorrelation Coefficient (ρ)

After identifying potential autocorrelation in residuals through the Durbin-Watson test, it was necessary to estimate the autocorrelation coefficient (ρ). This coefficient represents the relationship between an error term and its lagged value. To estimate it, a linear regression of the original model residuals (e_t) on their lagged values (e_{t-1}) was performed:

$$e_t = \rho e_{t-1} + u_t \tag{4}$$

where:

- e_t : current observation residual.

- e_{t-1} : previous observation residual.
- u_t : white noise error, independent and uncorrelated.

This regression corresponds to a **first-order autoregressive model (AR(1))**, also known as a **first-order Markov model**.

5.1.1 Results

From the estimated model, the coefficient ρ was found to be:

$$\rho = 0.2197 \quad (5)$$

indicating a moderate positive correlation between consecutive residuals of the original model.

5.1.2 Model Interpretation

The estimated model can be interpreted as an **AR(1) model**, which assumes that the current value of the residuals e_t depends linearly only on the immediately preceding value e_{t-1} , along with a random error term u_t . This type of model is commonly used to describe time processes with short-term memory.

5.1.3 Conclusion

The estimation of ρ is an essential step to correct autocorrelation in the original model. The estimated value $\rho = 0.2197$ will be used in the model transformation, as described in the next section.

5.2 Application of the P Matrix to the Model

To correct for autocorrelation in residuals identified in the initial estimation, a transformation based on the **P matrix** was applied. This transformation eliminates the linear dependence among consecutive residuals, ensuring that the OLS assumptions are met.

The **P matrix** is a square matrix of size $T \times T$ (where T represents the number of observations) and is constructed using the estimated autocorrelation coefficient ($\hat{\rho}$). Its structure directly reflects the correlation between consecutive residuals and is defined as follows:

- **Main diagonal:**
 - The first element is $\sqrt{1 - \hat{\rho}^2}$, while all other elements along the main diagonal are equal to 1.

- **Above and below the main diagonal:**

- The elements immediately above and below the main diagonal are equal to $-\hat{\rho}$, representing the correlation between consecutive residuals.

- **Other elements:**

- All other elements of the matrix are 0, reflecting the absence of dependence among non-consecutive residuals.

For example, for $T = 4$, the matrix P is:

$$P = \begin{bmatrix} \sqrt{1 - \hat{\rho}^2} & -\hat{\rho} & 0 & 0 \\ -\hat{\rho} & 1 & -\hat{\rho} & 0 \\ 0 & -\hat{\rho} & 1 & -\hat{\rho} \\ 0 & 0 & -\hat{\rho} & 1 \end{bmatrix}$$

The **P matrix** was applied to transform both the dependent variable (y) and independent variables (X) of the original model. By premultiplying the original model $y = X\beta + \varepsilon$ by P , we obtain the transformed model:

$$P \cdot y = P \cdot X \cdot \beta + P \cdot \varepsilon \quad (6)$$

5.2.1 New Estimation of the Transformed Model

After applying the transformation through the **P matrix**, the model was estimated using **Ordinary Least Squares (OLS)**, now applied to the transformed model.

$$\tilde{y} = \tilde{X}\beta + \tilde{\varepsilon} \quad (7)$$

Where:

- $\tilde{y}$: transformed dependent variable ($P \cdot y$)
- $\tilde{X}$: matrix of transformed independent variables ($P \cdot X$)
- $\tilde{\varepsilon}$: transformed residuals, which should no longer be correlated

The formula for estimating the coefficients in OLS remains unchanged in matrix algebra, but it now applies to the transformed model:

$$\hat{\beta} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \tilde{y} \quad (8)$$

This formula has been manually implemented in R as follows:

```
1 beta_star <- solve(t(X_transformed) %*% X_transformed) %*% t(X_
  transformed) %*% y_transformed
```

The estimated coefficients ($\hat{\beta}$) represent consistent and unbiased estimates, as the autocorrelation in residuals has been eliminated due to the transformation.

5.3 Verification of Autocorrelation Elimination

Before analyzing the coefficient estimates, it is crucial to verify that autocorrelation in residuals has indeed been eliminated. To this end, the **Durbin-Watson test** was applied again to the residuals of the transformed model.

1. Calculation of transformed residuals

```
1 resid_transformed <- residuals(model_transformed)
```

2. Application of the Durbin-Watson test

```
1 durbinWatsonTest(resid_transformed)
```

3. **Test result:** The test statistic value was found to be **2.436**, which is close to the ideal value of 2, indicating the absence of autocorrelation in the residuals.

4. **Comparison with critical thresholds:** Given that $n = 44$ observations and $k = 3$ independent variables, with a significance level of $\alpha = 0.05$, the estimated critical thresholds are:

- Lower bound (d_L): 1.39
- Upper bound (d_U): 1.65

Since:

$$d = 2.436 > d_U = 1.65$$

we can conclude that the transformed residuals no longer exhibit autocorrelation.

5.4 Conclusion

We can **accept the null hypothesis** H_0 , meaning that the residuals of the transformed model are no longer correlated.

This result contrasts sharply with the original model, where:

$$d = 1.49 \quad (d_L < d < d_U)$$

This value fell within the **indeterminate zone**, making it impossible to accept or reject the null hypothesis, thus indicating an ambiguous situation and the need for transformation.

The transformation successfully eliminated autocorrelation in the residuals, as confirmed by the Durbin-Watson test. Consequently, the estimated coefficients in the transformed model are now **efficient, consistent, and suitable for statistical inference**.

At this point, we can proceed with analyzing the estimated coefficients and evaluating the impact of independent variables on the gold price range.

6 Results

In this section, we present the results of the econometric model estimates, both in its original form and after applying the transformation to eliminate autocorrelation in the residuals. We will analyze the estimated coefficients, their statistical significance, and the economic implications of the results.

6.1 Preliminary Results (Before Correction)

We report here the coefficients obtained **before the transformation**, along with their p-values and interpretations.

Variable	Coefficient ($\hat{\beta}$)	p-value	Interpretation
Intercept	14.027	—	In the absence of other factors (dummy = "No", NFP deviation = 0), the price of gold moves by an average of 14.027 points during the NFP release.
Weekly Volatility (β_1)	0.435	0.067	Higher volatility in the current week seems to increase intraday volatility, but the effect is not statistically significant.
Geopolitical Dummy "Yes" (β_2)	-2.353	0.303	The presence of relevant geopolitical events seems to reduce volatility, but the effect is not statistically significant.
NFP Forecast Deviation (β_3)	1.252	0.460	A larger deviation between actual and forecasted NFP values seems to increase volatility, but the effect is not statistically significant.

Table 1: Estimated coefficients before the transformation.

The preliminary results indicate that, although the variables of interest seem to have a potential effect on intraday volatility, none of them is statistically significant. Moreover, the Durbin-Watson test highlighted the presence of positive autocorrelation in the residuals, compromising the validity and efficiency of the estimates.

6.2 Final Results (After Correction)

We now present the coefficients obtained **after the transformation**.

Variable	Coefficient ($\hat{\beta}$)	p-value	Interpretation
Intercept	14.038	0.005	Even in the transformed model, the intercept remains significant and represents the average movement in the absence of other factors.
Weekly Volatility (β_1)	0.430	0.086	The positive effect of weekly volatility is confirmed but remains non-significant.
Geopolitical Dummy "Yes" (β_2)	-2.342	0.296	The negative effect of the presence of geopolitical events persists, but it is still not significant.
NFP Forecast Deviation (β_3)	1.217	0.475	The effect of the NFP data remains non-significant.

Table 2: Estimated coefficients after the transformation.

After applying the transformation, the Durbin-Watson test ($d = 2.43$) confirmed the absence of autocorrelation in the residuals, making the estimates efficient and unbiased. Although the coefficients have remained stable in value, none of the independent variables has reached statistical significance ($p > 0.05$), except for the intercept. This suggests that the factors considered may not be sufficient to fully explain the intraday volatility of gold prices during NFP releases.

7 Discussion and Conclusions

The final results suggest that, while weekly volatility and economic shocks appear to play a role in explaining gold price movements, these effects are not statistically significant. The presence of autocorrelation in the residuals was successfully corrected, but the lack of significance in the coefficients indicates that additional variables or approaches may be necessary for a more comprehensive understanding of intraday gold volatility dynamics during NFP releases.

However, the intercept of the transformed model is significant, suggesting that, even in the absence of variations in independent variables and under "normal" conditions (`dummyRes = No`), the gold price exhibits a consistent average movement during NFP releases. This result may reflect an underlying level of volatility linked to the intrinsic importance of this macroeconomic event for global investors.

This study represents a first step toward building more robust quantitative models for financial market analysis.

8 Limitations of the Study

Although this study provides a quantitative analysis of gold volatility in response to Non-Farm Payrolls releases, it presents some limitations that should be considered:

1. **Limited time horizon** The analysis is based on a sample of 44 observations corresponding to the monthly NFP releases. While this dataset is adequate for a preliminary analysis, extending the time window could enhance the robustness of the results.
2. **Volatility modeling** The model employed uses a simple OLS regression to analyze the relationship between explanatory variables and gold volatility. However, more sophisticated models, such as GARCH specifications, could provide a more accurate representation of price movements in the minutes following economic releases.
3. **Exclusion of other market factors** The model considers only three explanatory variables (weekly average volatility, deviation between forecasted and actual NFP values, geopolitical events). However, other factors such as Federal Reserve monetary policy, market liquidity, or investor sentiment could significantly influence gold price reactions.
4. **Nonlinear effects and interactions** The analysis assumes a linear relationship between the independent variables and gold price volatility. In reality, the market's response to economic shocks may be nonlinear or dependent on interactions between variables that the current model does not capture.

Despite these limitations, this study represents a first step toward a deeper understanding of gold volatility dynamics during major macroeconomic events. Future research could address these limitations by expanding the dataset, adopting more advanced models, and considering a broader range of market factors.

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